

Thesis Proposal: Generating Privacy-Preserving Synthetic Healthcare Data with GANs vs. LLMs – Utility and Privacy Trade-offs

Introduction

Health data are often sensitive and strictly regulated (e.g., by GDPR in Europe and HIPAA in the U.S.), which makes data sharing and open research difficult. Patient records contain personal information that must be protected, yet there is a pressing need for large datasets to develop and validate AI models for healthcare. **Synthetic data** has emerged as a promising solution: it involves generating artificial records that statistically resemble real patient data but do not correspond to actual individuals, thus *potentially* bypassing privacy concerns \cite{Murtaza2023}. By providing realistic but artificial patient data, synthetic datasets can enable research and model training without risking patient confidentiality. Recent studies underscore that synthetic health data can augment limited real datasets and facilitate advances in diagnosis and treatment while reducing privacy risks \cite{Murtaza2023}. However, creating high-quality synthetic data that maintains *utility* (i.e., usefulness for analysis or model training) **and** preserves patient *privacy* is a non-trivial challenge.

In the machine learning community, two generative modeling paradigms have gained prominence for data synthesis: **Generative Adversarial Networks (GANs)** and **Large Language Models (LLMs)**. GANs, introduced in 2014, learn to generate realistic data by pitting a generator and discriminator network against each other. They have been widely applied to healthcare data (especially structured EHR data) to create synthetic patient records \cite{Choi2017, Murtaza2023}. More recently, transformer-based LLMs (pre-trained on vast corpora and fine-tuned for specific tasks) have shown surprising capability to generate synthetic tabular data when treated as a language modeling problem \cite{Miletic2024}. LLMs can, for example, treat each row of a medical dataset as a sentence and generate new "records" in textual form. Early evidence suggests that LLMs can capture complex high-dimensional dependencies out-of-the-box due to their powerful sequence modeling and attention mechanisms, potentially matching or surpassing GANs on data quality \cite{Miletic2024}. Unlike GANs, which often require specialized handling of different data types (e.g. one-hot encoding of categorical variables), LLMs can ingest heterogeneous data as text and model it in a unified way \cite{Miletic2024}. This flexibility makes LLMs an exciting alternative for synthetic data generation in healthcare.

Despite their promise, both approaches come with challenges. GANs are known to suffer from training instabilities (mode collapse, convergence difficulties) and may struggle with capturing all modes of a complex distribution, especially for multimodal healthcare data. LLMs, on the other hand, are extremely large models that can be resource-intensive to train and, importantly, **pose privacy risks of their own**: LLMs are prone to memorizing training data, which means a naïvely trained LLM could regurgitate sensitive patient information verbatim \cite{Nahid2024}. Therefore, simply generating data with an LLM does not guarantee privacy without additional precautions. In this context, our thesis will **compare synthetic data generation using GANs versus LLMs in the healthcare domain**, with a specific focus on the trade-offs

between data utility and privacy preservation. We aim to rigorously evaluate how well each approach can produce useful synthetic health data and how each can be adapted or constrained to protect patient privacy. The outcome will inform best practices for researchers and practitioners deciding between GAN-based or LLM-based synthetic data generation for healthcare applications.

Motivation

The motivation for this research stems from both the **limitations of traditional data anonymization** and the **opportunities offered by synthetic data** in light of modern privacy regulations:

- **Shortcomings of Anonymization:** Conventional approaches to de-identify or anonymize health data (such as removing identifiers or applying k -anonymity) often fail to guarantee privacy. High-dimensional patient records can still be re-identified by linking quasi-identifiers or through sophisticated attacks, as numerous studies have shown. For example, even when obvious identifiers are removed, combinations of demographic attributes or clinical codes can uniquely identify individuals. This has led to high-profile cases of privacy breaches and a recognition that de-identified data is not inherently safe. Moreover, aggressive anonymization (e.g., generalizing or suppressing data) tends to destroy data utility, reducing the value of the dataset for research. Thus, relying solely on traditional anonymization either leaves privacy risks or renders data nearly unusable. This status quo is untenable in a world where rich datasets are needed for machine learning, yet individual privacy must be respected.
- **Regulatory and Ethical Pressure:** Privacy regulations such as the **EU GDPR** and **US HIPAA** impose strict rules on using and sharing personal health information. Organizations face legal and ethical obligations to protect patient data. Under GDPR, for instance, improperly sharing data that can be traced back to individuals can lead to severe fines. These regulations have had a chilling effect on data sharing across hospitals, researchers, and industry, slowing down innovation. There is a clear need for methods that allow compliance with these regulations while still enabling the secondary use of health data (for research, quality improvement, etc.). Synthetic data offers a potential way forward: if the data are sufficiently *de-linked* from real individuals, they may not be considered personal data at all under certain regulatory frameworks. In principle, a well-crafted synthetic dataset could be shared freely, enabling collaboration and open research without violating privacy laws.
- **Synthetic Data for AI Development:** The healthcare AI community has increasingly recognized synthetic data as a key tool for overcoming data scarcity and privacy hurdles \cite{Murtaza2023}. Unlike encryption or federated learning (which allow computing on real data without sharing raw records), synthetic data creates an *artificial* dataset that can be openly distributed. This is powerful for several reasons: (1) **Data Augmentation:** Synthetic records can supplement small real datasets, addressing class imbalances or rare conditions. (2) **Benchmarking and Competition:** Researchers can release a synthetic version of a proprietary clinical dataset to organize machine learning competitions or benchmarks, spurring innovation on clinically relevant problems without disclosing real patient information. (3) **Algorithm Development and Testing:** Synthetic data can be used in early-stage development of models (for example, validating an ICU patient mortality prediction model on synthetic ICU data before deploying on real data). It lowers the barrier to entry for researchers who lack access to large clinical databases. In all these cases, the goal is to maintain

utility — the synthetic data should retain meaningful patterns and relationships so that models trained on it will behave similarly to models trained on real data.

- **Balancing Utility and Privacy:** Achieving a balance where synthetic data is both **useful** and **safe** is challenging. On one hand, we want the synthetic data to be as realistic as possible (high utility), meaning it reproduces the complex correlations and distributions of the real data. On the other hand, if synthetic data points are too close to real ones (e.g., if a particular patient's record were copied almost exactly), then privacy is compromised. There is an inherent tension: the more fidelity synthetic data has to real data, the higher the risk of revealing information about real individuals. This trade-off is at the heart of our research questions. Prior work suggests that with careful techniques, one can generate data that preserves aggregate statistics and ML performance while still providing strong privacy guarantees (for instance, through differential privacy) \cite{ADS-GAN2020, Nahid2024}. We are motivated to quantitatively explore this trade-off for GANs and for LLMs. Each approach may have different privacy leakage profiles — for example, GANs might be less prone to exact memorization but could still leak partial information, whereas LLMs might memorize rare training examples verbatim unless regularized. Understanding these differences is important for choosing the right tool for a given scenario. Moreover, investigating techniques like *differential privacy (DP)* in conjunction with GANs or LLMs can illuminate how to systematically control privacy loss. Regulations such as GDPR even recommend privacy-by-design approaches like data synthesis and differential privacy; thus, a rigorous study of how these techniques perform on real healthcare data will have practical implications for regulatory-compliant data sharing.

In summary, this thesis is motivated by the urgent need in healthcare for data sharing solutions that do not compromise patient privacy. Synthetic data generation using advanced generative models stands out as a promising approach. By comparing GANs and LLMs, we aim to identify a strategy (or a combination of strategies) that yields high-quality data for analysis and machine learning, while minimizing the risk of privacy violations. This will help healthcare institutions and researchers navigate regulatory hurdles and unlock data for beneficial uses, striking a nuanced balance between innovation and privacy.

Research Questions and Approach

This research will be guided by two primary research questions, each addressing a critical aspect of the utility-privacy trade-off in synthetic data generation. For each question, we outline the approach to investigate it:

RQ1: Utility – *How does the data utility of synthetic healthcare data generated by GAN-based models compare to that generated by LLM-based models?* In other words, when using GANs vs. LLMs to create synthetic patient data, which approach produces data that is more useful for downstream analytical or machine learning tasks? To answer this, we will **evaluate the fidelity and utility** of synthetic datasets produced by each method. Our methodology involves training various GAN models and LLMs on real healthcare datasets (see **Methodology** and **Chosen Models** below), then using the generated synthetic data in tasks such as predictive modeling. For example, we will train a classifier (e.g., a random forest or neural network) on synthetic data and evaluate its performance on real data (and vice versa) \cite{Choi2017, Miletic2024}. We will compare performance metrics like accuracy or AUROC as a measure of how well the synthetic data retained the patterns of the real data. Additionally, we will calculate statistical similarity metrics between real and synthetic datasets (distribution of variables, pairwise correlations, etc.) to quantitatively measure fidelity. By conducting these evaluations across multiple datasets and model variants, we will determine

whether GANs or LLMs yield higher utility synthetic data. Our approach will also examine scenarios such as limited data regimes (how do models perform when the training data is small?) and the effect of model size (for LLMs) or architecture complexity (for GANs) on utility. We hypothesize that LLM-generated synthetic data can achieve competitive or superior utility compared to GANs, based on recent evidence \cite{Miletic2024}, but this may depend on the specific context (e.g., type of data, model configuration).

RQ2: Privacy – *How do GAN-based and LLM-based synthetic data generation methods compare in terms of privacy preservation, and what are the trade-offs between privacy and utility for each?* This question focuses on the privacy aspect: we will assess the risk of privacy leakage in synthetic data from GANs vs. LLMs. Our approach includes conducting **membership inference attacks** and **attribute inference tests** on the synthetic datasets to evaluate privacy. In a membership inference attack, an adversary tries to determine if a given real individual's data was used in training the generative model; success at this task indicates privacy leakage. We will simulate such attacks on both GAN and LLM generators following known methodologies (e.g., shadow modeling or distance-based inference \cite{ADS-GAN2020}). For attribute inference, we will test whether an adversary who knows some attributes of a patient can guess missing attributes by leveraging the synthetic data (for instance, by finding the closest synthetic record matching the known attributes). Furthermore, we plan to integrate **differential privacy (DP)** mechanisms into both GAN and LLM training pipelines (see **Methodology**) to provide formal privacy guarantees (with privacy budget ϵ), and then evaluate how this impacts the utility of the synthetic data. By varying the strength of privacy protection (for example, tuning ϵ in DP-SGD training, or adjusting an identifiability loss term in a GAN \cite{ADS-GAN2020}), we will observe the resulting changes in data utility. This will allow us to plot privacy-utility trade-off curves for each approach. Comparing these curves for GANs vs. LLMs will answer which approach can achieve a better balance – e.g., does one method degrade in utility more gracefully than the other as privacy protection is tightened? We anticipate finding some differences; for instance, LLMs might require more careful regularization to prevent memorization, whereas GANs might inherently obscure individual data points but could suffer other privacy issues like mode collapse leaving out minority groups (which is a form of bias and utility loss). Ultimately, by addressing RQ2 we will provide guidance on the relative privacy robustness of GANs vs. LLMs and how to tune each for high privacy with acceptable utility.

In summary, **RQ1** looks at *how well each method reproduces the useful information* in the data, and **RQ2** examines *how well each method protects the sensitive information*. The methodology for both questions involves empirical experiments with multiple models and datasets, quantitative metrics, and where possible, statistical hypothesis testing to ensure any observed differences are significant. By explicitly tackling these two questions, our thesis will deliver a comprehensive comparison of GANs and LLMs for privacy-preserving synthetic data generation in healthcare.

Methodology

To address the research questions, we will follow an experimental methodology with several key components: **model selection**, **dataset preparation**, **training pipeline**, and **evaluation protocol**. We plan to conduct a rigorous side-by-side comparison of GAN-based and LLM-based synthetic data generators under similar conditions. The steps below outline our approach in detail.

Model Selection and Configuration

We will implement or utilize representative models from both the GAN family and LLM family, drawing from state-of-the-art examples in the literature:

- **GAN-Based Models:** We will start with **medGAN** \cite{Choi2017}, the seminal GAN for medical data that introduced an autoencoder architecture to handle high-dimensional binary EHR records. MedGAN will serve as a baseline for generating multi-label patient data (e.g., diagnoses). We will also use **CTGAN** \cite{Xu2019CTGAN} (Conditional Tabular GAN) for synthesizing mixed-type tabular data; CTGAN is a modern GAN variant designed to handle continuous and categorical variables with techniques like mode-specific normalization. Additionally, for a privacy-focused GAN, we will incorporate **ADS-GAN** (Anonymization through Data Synthesis GAN) \cite{ADS-GAN2020}. ADS-GAN is based on WGAN-GP (Wasserstein GAN with Gradient Penalty) with a conditional GAN setup, and it introduces an *identifiability loss* to penalize synthetic samples that are too close to real samples. This model explicitly tries to balance privacy and realism by enforcing a minimum distance between synthetic and real records \cite{ADS-GAN2020}. We will either implement ADS-GAN or adapt an existing WGAN-GP implementation by adding the identifiability constraint as described by Yoon et al. \cite{ADS-GAN2020}. If time permits, we may also consider **DP-CTGAN** or **PATE-GAN** as additional baselines: DP-CTGAN extends CTGAN with differential privacy \cite{Torfi2022}, and PATE-GAN uses the Private Aggregation of Teacher Ensembles framework to provide DP guarantees in GAN training. These would allow us to test *formal DP-guaranteed GANs* versus our other methods. However, our core GAN models will be medGAN (for binary clinical data) and CTGAN/ADS-GAN (for general tabular data), which cover both foundational and state-of-the-art techniques.
- **LLM-Based Models:** For the LLM approach, we will leverage transformer-based language models that can be trained or fine-tuned to generate synthetic data. We plan to use the open-source **Pythia LLM suite** (from EleutherAI) as used by Miletic and Sariyar \cite{Miletic2024}, which provides models of various sizes (e.g., 16 million, 160 million, up to 1 billion parameters). By choosing a range of model sizes, we can investigate the effect of model capacity on synthetic data quality (as larger models may capture more structure but also risk more memorization). Each model will be fine-tuned on our structured datasets by linearizing the records into a textual format. For example, a single patient record (with fields like age, diagnoses, lab values, outcome) can be serialized into a sentence or a JSON-like string which the LLM learns to generate. The LLM training objective will be to maximize the likelihood of real records (modeled as text sequences) so that it can sample new similar records. We will also consider a GPT-style model (such as GPT-2 or GPT-Neo) if needed, but Pythia covers this well. On the **privacy front** for LLMs, we will incorporate techniques from a recent study *SafeSynthDP* \cite{Nahid2024}: specifically, we will experiment with **differentially private fine-tuning** of the LLM via DP-SGD (stochastic gradient descent with noise added to gradients, providing an (ϵ, δ) -DP guarantee for the training process). This will limit the extent to which the LLM training can memorize any individual record. Additionally, we may try output perturbation methods: for instance, adding calibrated noise to the generated probabilities or sampling process such that the final output also satisfies differential privacy \cite{Nahid2024}. The goal is to have an LLM variant that is privacy-preserving by design, to compare against the likes of ADS-GAN or PATE-GAN. For clarity, we will likely have two versions of our LLM model in experiments: one non-DP LLM (to maximize utility and see worst-case privacy leakage) and one DP-enhanced LLM (to evaluate the privacy/utility trade-off).

All models will be configured and trained using consistent pipelines (as described below) to ensure fairness in comparison. Hyperparameters (learning rates, training epochs, etc.) will be tuned on hold-out validation sets for each model type. In the case of GANs, we will monitor training for convergence and use techniques like early stopping if mode collapse or divergence is detected. For LLMs, we will monitor training loss and use standard language model validation perplexity to avoid overfitting (especially critical when training with DP, which tends to noisify gradients and could underfit if over-regularized).

Datasets and Data Preparation

We will evaluate the models on **several healthcare datasets**, chosen to represent both *structured tabular data* and potentially *unstructured text data*. The rationale is to cover different types of health data commonly encountered:

- **MIMIC-III Clinical Database** \cite{Johnson2016}: MIMIC-III is a large, publicly available ICU (intensive care unit) database with de-identified health records for ~40,000 critical care patients. It contains a wealth of structured data: demographics, vital signs, lab measurements, diagnoses (ICD codes), procedures, medication events, and more. We will use a subset of MIMIC-III focusing on a specific prediction task. For instance, we might extract a cohort of ICU stays with features such as age, gender, a summary of comorbid diagnoses, and outcomes like in-hospital mortality or length-of-stay > 7 days (as a binary label). This structured dataset (perhaps with on the order of 10k samples and dozens to hundreds of features) will serve as a realistic testbed for synthetic data generation. It has mixed data types (continuous vitals, categorical codes, binary flags) and class imbalance (mortality may be ~10%), providing challenges for generative models. We will apply common preprocessing: e.g., one-hot encode categorical codes or group them by categories (for GAN inputs), normalize continuous features, and filter out any personal identifiers (MIMIC is already de-identified). For LLM training, we will convert each patient record into a textual summary format (e.g., "Age: 65; Gender: F; Diagnoses: [Heart failure, Diabetes]; ICULOS: 9 days; Outcome: Mortality=Yes."). We will ensure this textual format is consistent and parseable so that generated sequences can be translated back into structured form for evaluation.
- **UCI Heart Disease Dataset** \cite{Dua2019}: This is a smaller tabular dataset (available from the UCI Machine Learning Repository) with about 300 patient instances and ~14 clinical features (age, sex, chest pain type, blood pressure, cholesterol, etc.) used to predict presence of heart disease. We include this dataset for a couple of reasons: it is a well-known benchmark in medical ML, and it allows testing our methods on a smaller-scale, purely tabular problem (with mainly numerical and categorical inputs). The small size is a stress-test for overfitting—GANs might struggle with so few samples, and LLMs might easily memorize them—so it will be interesting to see which method fares better. We will perform minimal preprocessing (since features are already numeric or categorical) and use it primarily for evaluating utility (since privacy is less of a concern with such a tiny dataset, though we can still simulate privacy metrics as a methodological exercise).
- **De-identified Clinical Text (Optional)**: In addition to structured data, we are considering an unstructured dataset such as a set of *de-identified clinical notes* (e.g., discharge summaries from MIMIC-III, which have had PHI removed). This would allow us to explore LLM-generated synthetic *textual* data, like synthetic clinical narratives. However, generating realistic clinical text with GANs is impractical (GANs for text are not well-developed), so this comparison would mainly evaluate the LLM's ability to produce useful text and assess privacy in text generation. This is a secondary

objective; our primary focus will remain on structured/tabular data where both GAN and LLM can be applied. If included, we would use a sample of, say, 1,000 ICU discharge summaries, train an LLM to produce notes, and then evaluate whether the synthetic notes preserve medical semantic content while concealing patient identity. Privacy evaluation here might involve checking that no real patient names or exact phrases appear in the synthetic notes. We list this dataset as optional/extension, to be attempted if time allows, since it broadens the scope beyond direct GAN-vs-LLM comparison.

For each dataset, we will create a **training set**, a **validation set**, and a **hold-out test set of real data**. The generative models (GANs or LLMs) will be trained on the training set. The hold-out real test set will never be seen during generation; it will be used to evaluate utility (e.g., training a model on synthetic data and testing on this real set, or vice versa). The separation of data is also important for privacy tests: membership inference attacks will treat the training set members vs. hold-out individuals differently. We will ensure any hyperparameter tuning is done on the validation set to avoid leaking test information. In summary, careful dataset handling will enable unbiased evaluation of both utility and privacy for the synthetic data generators.

Training Pipeline

The training pipeline will differ slightly for GANs vs. LLMs, but we will keep the overall process parallel:

- **GAN Training:** Using the prepared training data, we will train each GAN variant (medGAN, CTGAN, ADS-GAN, etc.) to learn the distribution of the data. For medGAN (which was originally designed for binary multi-hot vectors of diagnoses), we will use a variant if needed to accommodate continuous features (or bin continuous features into categories to apply medGAN, though CTGAN is more natural for continuous). CTGAN and ADS-GAN will be trained on mixed data directly; CTGAN uses conditional vector sampling to handle categorical variables which we will set up. We will use batch training with appropriate batch sizes (experiments will determine a stable size, perhaps 128 or 256). The loss functions: medGAN uses a standard GAN loss plus reconstruction loss from the autoencoder; CTGAN uses a logit loss for discrete columns and WGAN objectives for continuous; ADS-GAN uses the WGAN-GP loss with an additional identifiability loss term [ADS-GAN2020]. We will implement the identifiability loss by periodically measuring the distance (e.g., Euclidean or some normalized distance) between each synthetic sample in a batch and its nearest real sample, then adding a penalty if this distance is below a threshold. The gradient of this penalty will push synthetic data away from real data points, thereby reducing overfitting to individual records. We will train GANs for a sufficient number of epochs (possibly on the order of a few hundred, depending on convergence). Throughout training, we will monitor generator loss, discriminator loss, and generated sample quality (by checking basic statistics match) to decide when to stop. If mode collapse is observed, we will employ techniques like restarting with different random seeds, tuning the learning rate, or using historical average discrimination. For ADS-GAN specifically, a hyperparameter for the identifiability constraint (how heavily to weight it) will be tuned: a higher weight should improve privacy (synthetics far from reals) but may harm fidelity; we will experiment to find a balanced setting or perhaps multiple settings (to produce curves for trade-off analysis).
- **LLM Training:** The LLM (transformer) models will be fine-tuned on the same training data, but in text form. The training process is essentially next-token prediction on sequences that represent one or multiple records. We will likely format each record as a single sequence terminated by a special token (e.g., `<EOR>` for end-of-record). The model will learn to emit valid records token by token.

Fine-tuning will be done using a language modeling loss (cross-entropy). We will use libraries such as Hugging Face Transformers to load pre-trained weights for Pythia models and then train on our corpus of records. A key difference from GANs is that LLM training is not adversarial; it's a maximum-likelihood training which tends to be more stable (no generator-discriminator fighting). However, as noted, overfitting is a concern, especially with smaller datasets. We will apply regularization techniques: early stopping based on validation perplexity, possibly dropout in the model, and crucially, **differential privacy during training** for certain runs. When training with DP-SGD, we will use a tool like Opacus (for PyTorch) to clip gradients and add Gaussian noise. We will target a range of privacy budgets, e.g., $\epsilon = 1, 5, 10$ (with $\delta = 10^{-5}$) to see how it affects model training. The DP training will generally reduce the effective capacity of the model (due to noise), so we anticipate needing more training epochs or a smaller model for DP to converge. In non-DP LLM training, we will train until the model starts to memorize (detected by training loss dropping much lower than validation loss). We may also incorporate *prompting or conditioning*: for instance, instruct the LLM to generate a record given some prompt (like "Generate a new patient record:"). But in our simplest setup, we will have the model generate records unconditionally by sampling from it repeatedly (each sample tends to produce one record and then stops at EOR token).

- **Differential Privacy Integration:** As mentioned, we plan to integrate DP in both paradigms for certain experiments. For GANs, one approach is **DP-SGD on the discriminator** (which is a known method to make GAN training differentially private, by clipping and noising the discriminator's gradient updates) \cite{Xie2018DPGAN}. Another approach is the PATE framework (training multiple teacher discriminators on disjoint subsets and using an aggregated student) \cite{Jordon2018PATEGAN}, but that might be complex to implement. We will likely implement the former: use an optimizer on the discriminator with per-sample gradient clipping and noise addition to guarantee DP. The generator updates in that case do not get direct access to real data except through the discriminator, so the whole model can be shown to satisfy DP. We will compare this DP-GAN training to the ADS-GAN approach (which is not formally DP but provides heuristic privacy). For LLMs, as described, DP-SGD will be our main tool. Alternatively (or additionally), **output perturbation** can be considered: after training a model, when generating synthetic data, one can probabilistically omit or alter some rare details to reduce the chance of replicating a unique training record (like not outputting sequences that are too close to any training sample). The *identifiability metric* from ADS-GAN can even be applied post-hoc: compute distances from each synthetic record to all real training records, and if any synthetic is uncomfortably close to a real, we could discard or tweak it. This hybrid approach (generate freely then filter using a privacy metric) will be explored if time allows, as it could be applied to both GAN and LLM outputs for fairness.
- **Statistical Validation:** Throughout training and generation, we will apply statistical validation checks. For example, after generating a synthetic dataset, we will compare summary statistics to the real data: means, variances, correlations, and distributions of key features. We will use two-sample hypothesis tests (Kolmogorov-Smirnov for continuous distributions, chi-square tests for categorical distributions) to see if differences are significant. This will help us gauge if a model has underfit or overfit. If we find a model yields significantly different distributions on critical variables, we may adjust it (e.g., tweak the loss or add a constraint). These statistical checks ensure that the synthetic data *looks realistic* before we proceed to formal utility evaluation. They also feed into our utility metrics later.

The entire pipeline will likely be executed multiple times for different models and settings. For reproducibility, we will fix random seeds for each run and document the configurations. We will also likely use **cross-validation or multiple random splits** to ensure that results are not due to a particular train-test split idiosyncrasy. This means training each model perhaps 3-5 times with different splits or seeds and averaging performance; this will give more robust comparisons between GAN and LLM results.

A simplified diagram of the planned pipeline is illustrated in Figure \ref{fig:overview}. It depicts how real data is used to train either a GAN or an LLM (with optional differential privacy), how those models generate synthetic data, and how the synthetic data is then evaluated via utility tasks and privacy attack tests.

\begin{figure}[htbp] \centering \includegraphics[width=0.9\textwidth]{figures/architecture.png} \caption{Overview of the methodology. Real healthcare data (training set) is used to train a GAN model and an LLM model in parallel. The GAN training may include an identifiability loss or DP-SGD on the discriminator for privacy, while the LLM training may include differential privacy in gradient updates. Once trained, each model generates synthetic data. The synthetic data is then assessed through evaluation modules: a utility evaluation (e.g., train classifier on synthetic, test on real) and a privacy evaluation (e.g., membership inference attack using real vs. synthetic data). The outcomes (utility metrics and privacy risk metrics) are compared to analyze trade-offs between data utility and privacy for GANs vs. LLMs.} \label{fig:overview} \end{figure}

Evaluation Strategy

Once we have trained models and produced synthetic datasets, we will perform a comprehensive evaluation along two dimensions: **utility** (how useful/accurate the synthetic data is for analysis) and **privacy** (how well individual information is protected).

Utility Evaluation: We will employ several metrics and tests to quantify the utility of synthetic data: - *Machine Learning Efficacy:* The primary measure will be the performance of predictive models on downstream tasks using synthetic data. For each dataset/task (e.g., mortality prediction in ICU, or heart disease diagnosis), we will train a classifier on the synthetic dataset and evaluate it on a real-data test set (that was held out from training). We will compare this performance to a benchmark: the performance of the same model trained on the real training data. If synthetic data is high utility, the gap in performance should be small. For example, if a logistic regression trained on real ICU data achieves 0.80 AUC on the test set, we hope a logistic regression trained on synthetic ICU data might achieve a similar AUC (say 0.75-0.79). We will report metrics like accuracy, AUC, F1-score, depending on the task. This approach has been used in many prior works \cite{Choi2017, Miletic2024, ADS-GAN2020} as a way to assess whether synthetic data retains useful signal. We will do this for each generative model (GANs and LLMs) and compare. Furthermore, we can do the reverse experiment: train on real, test on synthetic. This checks if synthetic data covers the real data distribution (if a model trained on real data performs well in predicting labels on synthetic records, it implies synthetic records are drawn from a similar distribution). High bidirectional performance indicates the synthetic data is not only good for modeling but also *statistically similar* to real data.

- *Statistical Similarity Metrics:* We will calculate distributional similarity metrics between real and synthetic datasets. For each continuous feature, we can compute the Jensen-Shannon divergence or Wasserstein distance between the real and synthetic value distributions. For categorical features, we will compare category frequencies. We will also examine pairwise correlations (or higher-order if

needed) among features in real vs. synthetic. Another useful technique is dimensionality reduction visualizations: e.g., apply PCA or t-SNE on the combined real+synthetic dataset and see if synthetic points overlap real points or form separate clusters. If synthetic data is truly representative, we expect overlap in these projections. We may quantify this by measures like coverage (what percentage of real data points have a synthetic neighbor within a certain distance in feature space, and vice versa). Prior studies often report such statistical measures to complement ML task results \cite{Murtaza2023, Hernandez2022}.

- *Domain-Specific Validity:* For some datasets, we can include checks for logical consistency or clinical validity. For instance, in the synthetic data we should not see impossible combinations (e.g., a pregnant male patient, or a lab value outside human range). While our models won't inherently guarantee such constraints, if we incorporate them (some advanced GANs penalize invalid combinations \cite{Xu2019CTGAN, Murtaza2023}), we should verify them. We can define a set of rules (perhaps with clinician input or from data dictionaries) and verify synthetic data against them. The fraction of records violating any rule can indicate realism issues.

Each of these utility assessments will be applied to data from GAN models and LLM models. We will then be able to answer RQ1 by comparing results: for example, do LLM-generated synthetics yield higher classification accuracy than GAN-generated? Are the statistical divergences lower for one or the other? Because we are using multiple measures, we might see mixed results (e.g., GAN might match marginal distributions better, but LLM yields better predictive performance). We will interpret such differences to understand strengths/weaknesses: GANs explicitly optimize for distribution matching via the discriminator, while LLMs optimize likelihood, so their output might trade off different aspects of the data (GAN might better capture numeric distributions while LLM might better capture complex dependencies, as hypothesized in \cite{Miletic2024}). If needed, we will conduct **statistical tests** (like a t-test or Wilcoxon test on the differences in model performance across multiple runs) to establish significance of any observed advantages.

Privacy Evaluation: To assess privacy, we will simulate adversarial scenarios and measure information leakage: - *Membership Inference Attacks (MIA):* We will follow a standard evaluation procedure for MIAs \cite{Shokri2017}. First, we assume an adversary who knows the synthetic dataset and possibly has some auxiliary information. One approach is the distance-based MIA: for each real record in the training set and each real record not in the training set (but from the same population), the adversary finds the closest synthetic record (in terms of some distance metric). If that distance is below a certain threshold, the adversary guesses that the real record was in the training data used by the generator. We can vary the threshold and compute true/false positive rates, ultimately measuring the AUC of this membership classifier. A more learning-based MIA involves training a classifier on a bunch of sample records labeled as "training member" or "not a member" based on features that might be derived from the generative model's output (for example, using the generative model to compute log-likelihood of the record, since models often assign higher likelihood to seen training points). We will implement one or more of these strategies for both GAN and LLM outputs. Success of MIA is quantified by how much better than random chance the adversary can identify training members. If our synthetic data is perfectly private, an adversary should do no better than 50% in a balanced setting. We expect non-DP models might have some vulnerability (some AUC > 0.5). We will compare the MIA results of GAN vs LLM: e.g., does the LLM have higher membership leakage (perhaps because it memorized some entries) than the GAN? The literature suggests that over-parameterized models (LLMs) can memorize rare training data \cite{Carlini2021}, whereas GANs, by the nature of adversarial training, might *implicitly* protect against verbatim copying to some extent (though not

guaranteed). We will also evaluate how adding DP or identifiability loss changes the MIA success. Ideally, DP-trained models should show near random-chance MIA performance, confirming the privacy guarantee \cite{Nahid2024}. This will validate our privacy-preservation methods.

- *Attribute Inference Attacks:* In an attribute inference (also called attribute disclosure) scenario, the adversary knows some attributes of a target individual (e.g., they know a patient's age and diagnosis) and wants to infer a sensitive attribute (e.g., HIV status) by using the synthetic data. We will simulate this by randomly picking some records and splitting their attributes into observed vs secret. The adversary will then look at the synthetic dataset for records that match or are very close to the known attributes, and then attempt to guess the secret attribute from those matches (for example, if 90% of matching synthetic records have HIV=Yes, the adversary would infer Yes). We will measure the accuracy of such guesses compared to the true secret attributes of the real individuals. If synthetic data is private, it should not enable an adversary to do significantly better than baseline guessing. We might evaluate this under various conditions (e.g., how many attributes are known; we might follow the approach of Choi et al. \cite{Choi2017} who simulated an adversary with partial knowledge to see if synthetic data gave away the rest). The attribute inference risk will be reported as something like: "Given X attributes, the probability of correctly inferring attribute Y using synthetic data was P%, compared to baseline Q% without synthetic data." We will evaluate this for GAN-synthesized vs LLM-synthesized data. A model that memorizes data would allow near-perfect inference for some records, which would be a red flag.
- *Identifiability Measure:* In addition to formal attacks, we will use the **identifiability metric** proposed in ADS-GAN \cite{ADS-GAN2020}. This involves computing the distance of each synthetic record to the nearest real record (in the training set). We can plot the distribution of these nearest-neighbor distances for different models. If a model occasionally produces a synthetic that is almost identical to a real record, we will see some distances near 0. A privacy-safe generative model should produce a distribution of distances that has a reasonably large minimum (no exact matches, and hopefully no extremely close matches). We will compare these distributions: e.g., perhaps a vanilla GAN might have a smaller minimum distance (riskier) whereas ADS-GAN or DP-LSTM will push this minimum distance up. This metric provides an intuitive sense of privacy: larger nearest-neighbor distances mean each synthetic person is sufficiently different from any real person. We will decide on a threshold for "identifiability" (for instance, a distance below τ counts as potentially identifying someone) and compute the fraction of synthetic records that are too close. This complements the MIA results by directly measuring record similarity.

All privacy evaluations will be carried out under controlled conditions and repeated for each model. We will pay attention not just to average-case but worst-case leakage. For example, even if most records are safe, a single extreme outlier (like a synthetic record that exactly matches a rare real patient) would be problematic; we will report if such cases occur. If we find any, we will analyze why (it could indicate overfitting by the model) and note how our DP interventions mitigate it.

Trade-off Analysis: With utility and privacy metrics in hand, we will produce a comparative analysis. A key part will be articulating the **trade-offs**. For each approach (GAN, LLM), we will show how utility changes as we increase privacy protection. For instance, using differential privacy, we will have runs at different ϵ levels; using ADS-GAN identifiability, we could vary the weight of that loss. These will yield points on a curve (privacy vs. utility). We can plot, say, classification accuracy (utility) against membership inference success rate or against ϵ (privacy level). This visualization will help determine which model is

inherently more efficient in the trade-off. Perhaps an LLM might achieve a given utility at a lower privacy cost than a GAN, or vice versa. We will also consider the case of “no explicit privacy” (just the model’s natural state) to see baseline performance vs. privacy. Interestingly, it might turn out that even without DP, one method inherently leaks less. Or it might be that without any privacy measures, both leak too much, underlining the necessity of privacy-by-design.

We will use these analyses to answer RQ2 fully. We also plan to validate the privacy findings with theoretical expectations: for example, if we achieve an $(\epsilon=1)$ -DP guarantee in training the LLM, theoretically no single record should have a large influence; thus membership inference should be essentially null. If our empirical attack finds anything otherwise, that could be due to the gap between theoretical worst-case and practical average-case or due to implementation issues. We will reconcile these points.

Finally, beyond numeric evaluation, we may qualitatively inspect a few synthetic records to illustrate what the models are doing. We will include examples (ensuring they are indeed synthetic and not memorized real ones) to show, for instance: a real patient vs. a synthetic patient profile side by side, to demonstrate realism, and maybe highlight differences that occur due to privacy constraints (e.g., “the DP-synthesized patient looks average and avoids rare combinations that were present in real data”).

Through this multi-faceted evaluation strategy, we intend to draw robust conclusions about the performance of GANs vs. LLMs for synthetic data in healthcare, and provide insights into how to choose or design models depending on whether utility or privacy is the primary concern.

Literature Review

Research on synthetic data generation in healthcare has grown rapidly in recent years, producing a rich body of literature. Here we provide a synthesis of foundational and recent work, focusing on GAN-based and LLM-based methods for health data synthesis, their relative strengths/weaknesses, and how they have been evaluated in prior studies.

GAN-based Synthetic Data in Healthcare: The use of GANs for generating health data was kick-started by Choi et al. with **medGAN** \cite{Choi2017}. MedGAN was innovative in tackling the challenge of high-dimensional *binary* data (e.g., thousands of possible diagnosis codes for a patient) by incorporating an autoencoder to learn a lower-dimensional representation, which the GAN then learned to generate. This addressed the issue that vanilla GANs struggled with discrete multi-hot vectors. MedGAN demonstrated that synthetic patient records (specifically, diagnosis code vectors derived from ICU data) could capture clinical patterns well enough that models trained on synthetic diagnoses could approximate the performance of those trained on real data \cite{Choi2017}. However, medGAN and its early successors largely focused on *utility/realism* and paid less attention to formal privacy guarantees. Choi et al. argued that since the GAN does not explicitly copy real records (there is “no direct mapping” between real and synthetic examples), it provides some inherent privacy \cite{Choi2017}. Subsequent research showed this assumption needed to be validated, as generative models can still leak information.

MedGAN’s success spurred numerous **variants and improvements**. For example, medWGAN and medBGAN replaced the vanilla GAN loss with Wasserstein loss or used boundary-seeking techniques to improve training stability \cite{Murtaza2023}. **HealthGAN** \cite{Yale2019} applied a similar concept to generate health records with WGAN-GP and showed improved univariate distribution matching over medGAN, while noting medGAN’s limitation in capturing multivariate structure. Many models attempted to

extend synthetic generation beyond binary data: some targeted continuous features, others multi-modal data. A consistent challenge identified in reviews is that early GAN models (mostly based on MLP architectures) didn't capture temporal dependencies in longitudinal records \cite{Murtaza2023}. Torfi et al. argued that recurrent or convolutional architectures are needed for sequential EHR data, as MLP-based GANs miss the temporal dimension \cite{Torfi2022}.

Recently, attention has turned towards **longitudinal EHR synthesis**. For instance, **Dual Adversarial Autoencoder (DAAE)** by Lee et al. \cite{Lee2020} generates sequential health records by combining a recurrent autoencoder with dual discriminators (one on the data space, one on latent space). It achieved better predictive modeling performance than medGAN on longitudinal data and was favorably reviewed by medical experts for realism. Another approach, **SynTEG** by Zhang et al. \cite{Zhang2021SynTEG}, specifically tackled generating sequences of visits (with diagnoses over time) using a conditional WGAN; it introduced a two-stage process to model time between visits and visit details, improving realism for multiple-visit synthetic patients. These works indicate that GANs can be adapted to more complex data structures, though often at the cost of more complex architectures and training.

Privacy-Preserving GANs: Given the sensitivity of health data, researchers have proposed modifications to GANs to enhance privacy. One notable framework is **PATE-GAN** (Jordon et al. 2018) \cite{Jordon2018PATEGAN}, which applied the Private Aggregation of Teacher Ensembles (PATE) approach to GANs. The idea is to train multiple “teacher” discriminators on disjoint subsets of data and then train a “student” generator using aggregated teacher knowledge with added noise, yielding a differentially private guarantee. PATE-GAN was an early method to formally bound privacy loss while generating data. However, it can be complex and sometimes degrade utility due to the noise.

Another line of work integrated DP directly into GAN training via gradient perturbation. Xie et al. (2018) introduced DP-GAN, which perturbs discriminator gradients during training to satisfy DP \cite{Xie2018DPGAN}. **ADS-GAN** (Yoon et al. 2020) \cite{ADS-GAN2020}, as introduced earlier, took a slightly different approach: instead of strict DP, it introduced an *identifiability loss* to penalize synthetic samples that are too close to any real sample. By including this term in the generator's loss, ADS-GAN actively discourages memorization of individual data points. The authors demonstrated that ADS-GAN achieved **better privacy** (measured by low re-identification risk and outperforming PATE-GAN in certain metrics) while still maintaining **high utility** (the synthetic data preserved statistical distributions and yielded predictive performance comparable to real data) \cite{ADS-GAN2020}. This result is encouraging: it suggests that carefully designed regularization can make GANs privacy-aware without completely reverting to DP's heavy noise addition.

Building on tabular GAN successes like **CTGAN** \cite{Xu2019CTGAN}, which is now a popular baseline for tabular data synthesis, researchers have tried to combine these with privacy measures. **DP-CTGAN** \cite{Fang2020} incorporated differential privacy into CTGAN for medical data. Fang et al. reported that DP-CTGAN's synthetic data had superior utility for classification tasks compared to some other DP baselines, although the absolute utility still dropped at very strong privacy levels. There is often a pattern: as privacy guarantees strengthen (lower ϵ), the synthetic data's fidelity worsens, highlighting the trade-off. Murtaza et al.'s comprehensive review \cite{Murtaza2023} notes that many GAN-based studies either do *utility evaluation only* or include a cursory privacy evaluation (if any). There is a need for unified evaluation: they found no single metric or framework was standard, making comparisons difficult across studies \cite{Murtaza2023}. Our work is in part motivated by that gap: we attempt a side-by-side evaluation under common metrics.

Other Generative Approaches (Non-GAN): It's worth noting that GANs are not the only generative models in use. Variational Autoencoders (VAEs) and Bayesian networks have also been applied to synthetic health data. For example, **VAE-based models** like medVAE or conditional VAEs have generated synthetic patient data with some success, often yielding more diverse outputs but sometimes less sharp fidelity compared to GANs. An example is **EVA** (Biswal et al. 2021) which used a conditional VAE for longitudinal EHR and reported good utility \cite{Biswal2021}. Meanwhile, from a more classical perspective, **PrivBayes** (Zhang et al. 2017) \cite{Zhang2017PrivBayes} used Bayesian networks with differential privacy to generate synthetic data and was a winner in a NIST competition. Its limitation was handling very high-dimensional data due to the curse of dimensionality in graphical models. The 2021 **PrivSyn** method \cite{Zhang2021PrivSyn} took a different approach by generating synthetic tabular data using *marginal distributions* and achieved DP; it essentially constructs a probabilistic model of low-dimensional marginals and samples from it. PrivSyn was shown to handle fairly wide datasets (100+ attributes) under DP with better utility than prior DP methods, by avoiding strong independence assumptions. These non-neural approaches (PrivBayes, PrivSyn) are important as they sometimes outperform deep models under strict privacy, but they may struggle to capture complex correlations that deep networks can.

One very different approach in healthcare is using **knowledge-driven simulations or agent-based models** to generate data (for example, simulating patient trajectories through Markov models). **SynSys** \cite{SynSys2019} by Dahmen and Cook is a case in point: rather than using a deep neural network, SynSys built a synthetic data generation system for smart home healthcare datasets using hidden Markov models and regression models. The method captured daily activity patterns of seniors in a smart home and generated sensor event sequences that were statistically similar to real ones. SynSys improved activity recognition accuracy in scenarios where only a small amount of real data was available \cite{SynSys2019}. This demonstrates that for certain types of data (time-series sensor data in this case), transparent generative models can suffice. However, HMMs and similar models might not scale or generalize well to very high-dimensional EHR data that include many heterogeneous variables. Still, SynSys is a reminder that the *ultimate goal* is synthetic data utility and realism—GANs or LLMs are just means to that end, and in some cases simpler or more interpretable models might be advantageous.

LLM-based Synthetic Data: The use of large language models for synthetic **tabular** data is a very recent development. Miletic and Sariyar's 2023/2024 study \cite{Miletic2024} is one of the first systematic benchmarks of LLMs vs. GANs in this context. They used the Pythia LLMs to generate tabular health data (by treating rows as text) and compared against CTGAN as a reference. Their findings were striking: as LLM size increased (from 14M to 1B parameters), the quality of synthetic data improved, and even the smallest LLM produced data of comparable utility to the GAN \cite{Miletic2024}. In fact, larger LLMs outperformed the GAN in downstream classification tasks, suggesting that the transformer could capture complex patterns (especially with enough model capacity and data). They noted a positive correlation between training data size and LLM performance as well, which is intuitive (more data helps the large models). The advantages of the LLM approach include not needing special encoding for categorical data (the model just learns the textual representations) and potentially capturing long-range dependencies or rare combinations thanks to the expressiveness of the model. However, they also point out challenges: LLM training can be slower and more computationally expensive than GAN training for the same data, and handling numeric precision or enforcing constraints (like ranges) might require careful formatting or prompts.

Another frontier for LLMs is generating **textual data** like clinical notes or dialog. There has been work (e.g., by Kaiser et al. or others) using GPT-2 or GPT-3 to generate fake medical texts for data augmentation in NLP tasks. These show that LLMs can produce fluent clinical narratives that sometimes even fool human

evaluators, but the risk is that if the model was fine-tuned on actual patient notes, it might emit real patient details (a privacy issue). This brings up the intersection of LLMs and privacy: very recently, **Hasan Nahid et al. (2024)** introduced **SafeSynthDP** \cite{Nahid2024}, demonstrating a methodology to inject differential privacy into LLM-based data generation. They fine-tuned GPT-2 on some data with DP-SGD and/or added noise to the generation process, and evaluated both utility (model performance on tasks using the synthetic data) and privacy (resilience to MI attacks). They concluded that it is indeed possible to find a "viable balance" where the synthetic data is useful yet the LLM does not leak individual training examples \cite{Nahid2024}. Specifically, their DP-enhanced LLM still allowed training of downstream models with only a minor accuracy drop, while membership inference attack success was near chance and sensitive details could not be recovered. This is an encouraging result for the LLM side, indicating that techniques exist to tame the privacy risks of language models.

Comparative Insights: Comparing GANs and LLMs, a few themes emerge in the literature: - **Complexity vs. Stability:** GANs often require more hand-holding in training (tuning losses, monitoring for mode collapse). LLMs require heavy compute but training is a direct likelihood maximization, which is generally stable given enough data. Once trained, GANs generate samples very quickly (feedforward pass), whereas LLM generation is sequential and can be slower (though still reasonably fast with smaller models on modern hardware). If generating millions of records, GANs might have an edge in speed. - **Handling Data Types:** GANs typically need explicit handling of different data types (binary, categorical, continuous) often through tricks like one-hot encoding, normalization, or customized loss per type. LLMs convert everything to text and rely on the model to implicitly learn types and formats, which is both a pro and a con: it's flexible, but can lead to formatting errors or nonsensical outputs if not properly constrained. For example, an LLM might generate an age of 200 or a nonexistent lab test unless it learned the distribution well or is prompted with schema constraints. - **Memory and Mode Coverage:** There is an argument that GANs, because of their two-player training, might focus on the most salient modes of the data and could miss outliers or minorities (mode dropping), whereas an LLM trained to maximize likelihood will try to model the entire distribution (including capturing those tails better). Murtaza et al. \cite{Murtaza2023} highlighted that many GAN studies lacked focus on *minority representation* or *long-tail distributions*. An LLM might naturally do better here, but at risk of memorizing those exact outliers. - **Evaluation Practices:** Literature shows a variety of evaluation strategies. Many early works (medGAN and variants) used statistical similarity metrics (like distance measures, PCA plots) and possibly simple predictive tasks \cite{Choi2017}. Later works (e.g., ADS-GAN) included explicit privacy metrics (membership/attribute disclosure simulations) in addition to utility metrics \cite{ADS-GAN2020}. The LLM studies use similar metrics now (classification accuracy, etc.), and SafeSynthDP adds formal attack evaluations \cite{Nahid2024}. Yet, there is no single agreed-upon metric of "synthetic data quality". Each study tends to report a set of metrics; for instance, some report *accuracy* and *coverage* (the fraction of real data distributions covered by synthetic samples within some tolerance), others report *Propensity Score* mixtures (to see if a classifier can distinguish real vs synthetic, the idea being if it cannot, they are indistinguishable distributions). In one survey, Hernandez et al. (2022) lament that there's "no universal method or metric to evaluate and benchmark" different approaches, and stressed the need for more generalizability and consistent evaluation frameworks \cite{Hernandez2022}. Our work tries to heed that by using multiple metrics and reporting them side-by-side for GANs and LLMs. - **Results from Key Studies:** To mention a few results: Beaulieu-Jones et al. (2019) created a GAN for sharing patient data and found that models trained on synthetic data could achieve 0.9+ of the real AUC in tasks, showing promise for utility \cite{BeaulieuJones2019}. They also found that adding noise (differential privacy) reduced performance somewhat but still kept models viable. Jordon et al. (2018) with PATE-GAN showed it was possible to get close to real data accuracy with ϵ as low as about 1 or 2 in some cases, but not all tasks. Yoon et al. (2020) with ADS-GAN showed essentially no significant drop in predictive utility

compared to non-private GAN, while dramatically reducing identifiability risk \cite{ADS-GAN2020}. On the LLM side, as noted, Miletic & Sariyar (2024) saw even a 14M parameter LLM can rival CTGAN, and bigger ones exceed it \cite{Miletic2024}. These suggest that both approaches are viable and that the gap in utility might be closing, with LLMs a strong new contender.

In summary, the literature indicates that **GAN-based methods** have a strong track record in generating structured health data with good utility, and recent variants address privacy with techniques like DP and identifiability constraints. **LLM-based methods** are emerging and show great promise, achieving high utility and with potential to integrate privacy as well. However, direct comparisons are few so far, which is precisely the gap this thesis aims to fill. Our work will contribute to the literature by providing a systematic head-to-head evaluation of a state-of-the-art GAN and an LLM on the same data and metrics. Additionally, by evaluating privacy in depth for both, we extend prior comparisons that mostly focused on utility. We hope our findings will synthesize and build upon works like ADS-GAN \cite{ADS-GAN2020}, SynSys \cite{SynSys2019}, Murtaza et al.'s review \cite{Murtaza2023}, and the new LLM studies \cite{Miletic2024, Nahid2024}, offering a clearer picture of where the field stands and where to go next.

Related and Prior Work

Several published works form the foundation and inspiration for this thesis. We briefly highlight key contributions from existing research that inform our approach and underline the significance of our proposed study:

- **ADS-GAN (Yoon et al. 2020)** \cite{ADS-GAN2020}: This work introduced a GAN model specifically aimed at data anonymization through synthesis. By incorporating an identifiability loss term, ADS-GAN was able to limit how closely synthetic samples could mimic real individuals. The key contribution was demonstrating that one can achieve *comparable predictive performance* to real data while significantly reducing re-identification risks. ADS-GAN serves as a strong baseline for privacy-preserving synthetic data; we will build on its concepts (e.g., identifiability metric) in designing our privacy evaluation. Our thesis extends ADS-GAN's idea by comparing it with an entirely different model class (LLMs) and by exploring formal DP guarantees alongside heuristic privacy measures.
- **medGAN (Choi et al. 2017)** \cite{Choi2017}: As one of the earliest models for synthetic EHR data, medGAN proved the viability of using generative models to create realistic high-dimensional health records. It introduced techniques to handle binary/categorical data in GANs. While medGAN itself did not deeply explore privacy, it set the stage for a flurry of research in synthetic health data. We use medGAN as a conceptual starting point and reference for how evaluation was done (Choi et al. performed both statistical similarity checks and a form of membership attack simulation to argue for privacy). Many subsequent works we cite (e.g., those in Murtaza et al.'s review) are variations or improvements of medGAN, so understanding medGAN's architecture and limitations guides us in choosing better GAN baselines.
- **Murtaza et al. (2023) – State-of-the-Art Review** \cite{Murtaza2023}: This comprehensive literature review aggregated findings from 70 studies on privacy-preserving medical data generation. It highlights common achievements and gaps: e.g., many studies show synthetic data can reach 85–95% of model performance of real data across various metrics, and synthetic data is effective for tasks like machine learning model training and software testing. Importantly, the review points out that evaluation metrics are inconsistent and that areas like longitudinal data and unified quality

metrics need more attention. This directly motivates our research design: we aim to evaluate multiple metrics (utility and privacy) side by side, and we include a longitudinal aspect by considering sequential data (if possible). The review also lists strengths and weaknesses of different approaches, which helps us identify which models to include (we selected models like CTGAN, ADS-GAN, etc., because these were noted as state-of-the-art or particularly relevant in the review).

- **Hernandez et al. (2022) – Systematic Review of Tabular Health Data Synthesis** \cite{Hernandez2022}: This survey, focusing on 2016–2021, similarly found that GANs dominate recent research and that no single model works best across all scenarios. They also emphasized generalizability issues — a model tuned for one dataset might not perform well on another. This informs our methodology: we plan to test multiple datasets (MIMIC-III, Heart Disease, etc.) and multiple models to see how generalizable each approach is. If a model only works well on one dataset, that will be apparent in our results. This systematic review also underscores the importance of comparing approaches on a level playing field, which is exactly what our thesis will do for GANs vs. LLMs.
- **Miletic & Sariyar (2024) – LLMs for Tabular Data** \cite{Miletic2024}: This very recent study is a cornerstone for the LLM side of our comparison. It is, to our knowledge, the first to rigorously show that LLMs can rival GANs for tabular health data. Their benchmark provides evidence that we cite to justify using the Pythia LLM family and to hypothesize LLMs’ strong performance. However, their work primarily looked at utility (classification performance) and did not delve into privacy. Our thesis will extend their findings by adding the privacy lens: for example, even if LLMs are more accurate, are they more prone to leaking data? This is a question they didn’t address but we will.
- **Hasan Nahid et al. (2024) – SafeSynthDP** \cite{Nahid2024}: This is a recent arXiv work that directly tackles large language models with differential privacy for synthetic data generation. It shows practically how to integrate DP and gives results on membership inference resistance. We view this as an existence proof that LLMs can be made privacy-preserving. We will draw on their methodology for our DP-LFM implementation. Additionally, their results (showing minimal utility loss with DP in some cases) set a benchmark that we hope to either replicate or compare against with our GAN models. If our GAN with DP performs similarly, that’s interesting; if one dominates, that yields insight. Essentially, SafeSynthDP informs our experimental setup for DP integration and provides baseline expectations.
- **SynSys (2019) – HMM-based Generation** \cite{SynSys2019}: While not the central focus of our thesis (since we focus on GAN vs LLM), SynSys is an example of a successful synthetic data project that did not use GANs. It reminds us that for certain structured problems, simpler models can suffice. We cite SynSys to highlight that synthetic data generation is a broad field and that alternative methods exist (HMMs, agent-based simulation, etc.). The inclusion of SynSys in our lit review is mainly to acknowledge the breadth of approaches and to emphasize why we zero in on GANs and LLMs (these are currently the two most powerful general-purpose paradigms, whereas SynSys was tailored to a specific sensor scenario).

In summary, these published works collectively paint the landscape in which our research sits. We are effectively combining insights from them: the privacy techniques of ADS-GAN, the evaluation thoroughness urged by reviews, the new capabilities of LLMs shown by Miletic et al., and the DP integration shown by Nahid et al. Our thesis will stand on the shoulders of these works but go beyond by directly comparing the

two paradigms and giving practical recommendations for privacy-preserving synthetic data generation in healthcare.

Chosen Datasets

We have selected a set of datasets that will allow us to evaluate synthetic data generation for both structured (tabular) and unstructured (text) healthcare data. Each dataset is chosen for its relevance, diversity in data type, and public availability (to ensure reproducibility):

- **MIMIC-III (Medical Information Mart for Intensive Care)** \cite{Johnson2016}: MIMIC-III is a large relational database containing de-identified health-related data associated with over 40,000 patients who stayed in critical care units of a single hospital. From MIMIC-III, we will derive one or more tabular datasets for our experiments. One likely choice is a table for an ICU mortality prediction task. For example, we can construct a dataset where each row represents a patient ICU stay with features such as demographics (age, gender), severity scores (like SAPS-II, if available), comorbid conditions (maybe a binary vector of common diagnoses), and outcome (death or survival). This gives a binary classification task which is common in clinical ML literature. The reason to choose MIMIC-III is that it's **real, complex clinical data** with rich distributions and correlations; if our models can synthesize MIMIC-like data well, that's a strong validation. MIMIC-III also contains sensitive information (even though de-identified, it has detailed health data), making it appropriate for testing privacy preservation. Another aspect is that MIMIC-III can be used to create different kinds of tables: we could try a purely numeric lab measurements table, or a diagnoses table (as medGAN did), etc., allowing us to test model performance on different data distributions all within one database. We will cite Johnson et al. (2016) for MIMIC-III and ensure compliance with its data use agreement.
- **UCI Heart Disease Dataset** \cite{Dua2019}: This is a smaller, well-known dataset (originally collected from Cleveland Clinic and other sites) with patient information and a binary label indicating presence of heart disease. It has around 303 instances and 14 attributes, including both numeric (e.g., cholesterol level) and categorical (e.g., exercise-induced angina yes/no) data. We include it to test our methods on a *small-scale* dataset that is prone to overfitting. If an LLM or GAN can generate decent synthetic data from only 300 records without just memorizing them, that would be impressive. On the other hand, it might reveal that certain models are not suitable for very small data (in which case one might not even attempt synthesis). The heart disease dataset is also straightforward for evaluating utility: a logistic regression or any classifier has a known baseline accuracy ~85% on this data, so we can see how close synthetic-data-trained models get. Additionally, since it's so small, we might attempt exhaustive privacy evaluation (e.g., checking if any synthetic record exactly matches a real one, which is feasible to do by brute force on 300 samples).
- **CDC Diabetes Health Indicators Dataset**: This dataset (available via UCI or Kaggle) was used by Miletic & Sariyar \cite{Miletic2024} in their LLM vs GAN comparison. It contains health survey data (possibly the Behavioral Risk Factor Surveillance System subset) with binary indicators for diabetes and health behaviors. If accessible, we will use it as a **second tabular health dataset**. It's larger than Heart Disease (on the order of tens of thousands of records) and mainly binary features, making it somewhat similar to medGAN's context. The advantage of including this dataset is twofold: (1) It allows us to directly compare our results with those reported by Miletic & Sariyar, since they might have metrics for CTGAN vs LLM on this data. (2) It adds variety in that it's not hospital data but survey data, which might have different distributions (perhaps a wide range of demographics, etc.).

If for some reason this dataset is not used, an alternative of similar nature could be the **Adult** dataset (though not healthcare, it's about income prediction with mixed data) or another public health dataset.

- **Clinical Text (MIMIC-III Notes):** We are considering a set of de-identified free-text clinical notes from MIMIC-III, such as discharge summaries or radiology reports. These are unstructured and require an LLM to generate. We would use this dataset in a limited capacity to test the LLM's text generation prowess and privacy. Each note could be quite lengthy, so we might sample paragraphs or sentences. The goal would be to see if a fine-tuned GPT-2 (for instance) on these notes can produce believable clinical text, and then check that it's not copying any original sentences (which we can check by search). We would not expect a GAN to handle this task, so this dataset might only be used to demonstrate LLM capabilities, not for the GAN vs LLM comparison. Due to time constraints, this may be beyond our core comparison, but it's listed as a possibility to show the generality of LLM synthetic data (structured vs unstructured).

Dataset Relevance: Each chosen dataset aligns with the types of data commonly encountered: - MIMIC-III covers **electronic health records (EHR)**, with structured clinical variables and outcomes, which is a primary use-case for synthetic data (to share EHR data for research). - UCI Heart Disease is a **small clinical study** dataset, representing scenarios where data is limited – can synthetic data augment it? - CDC Diabetes Indicators represents **population health survey** data, relevant for epidemiological studies and public health, where privacy is also crucial. - MIMIC notes represent **clinical text**, relevant for NLP applications like training clinical language models without real notes.

By using these, we ensure our findings are applicable across different sub-domains of healthcare: critical care, cardiology, general health, and clinical narrative. Moreover, all these datasets (except the clinical text, which is optional) are publicly available, meaning others can replicate or build on our work.

We will use appropriate preprocessing for each dataset as mentioned and document the exact versions and any pre-processing (for transparency). For example, if we one-hot encode ICD codes in MIMIC, we will state how many codes and how we selected them (perhaps top 100 frequent diagnoses to limit dimensionality, etc.). For continuous features, we might apply min-max scaling or z-scoring before feeding to models like GANs (since GANs do well with normalized inputs). For LLMs, we might bin continuous values into categories or strings (like "BloodPressureHigh", "Medium", "Low" categories) to avoid having too many numeric possibilities in text.

In summary, these chosen datasets will allow a thorough evaluation of our synthetic data generation approaches, covering different data characteristics (binary vs continuous, low-dimension vs high-dimension, small vs large sample size). Our experiments on them will illuminate how each generative method performs in varying conditions typical of healthcare data.

Chosen Models

We will utilize and customize a range of models in two categories – GAN-based and LLM-based – to generate synthetic data. Below is a list of the selected models, along with configuration details and rationale for each:

GAN-Based Models:

1. **medGAN (2017)** – *Medical Generative Adversarial Network* \cite{Choi2017}: We choose medGAN as a classic baseline. **Configuration:** We will use medGAN for binary multi-label data synthesis (e.g., diagnosis codes). It consists of an autoencoder (to compress and reconstruct real data) and a GAN (generator and discriminator) operating on the autoencoder's latent space. The generator and discriminator are feed-forward neural networks (MLPs). We'll implement medGAN closely following Choi et al.: the autoencoder will have a 128-dimensional latent space (tunable) and the generator will take random noise plus (optionally) class labels if we do class-conditional generation. Training involves alternating between autoencoder reconstruction (MSE loss) and GAN adversarial loss. **Rationale:** medGAN addresses discrete data specifically and is lightweight to train. It may not achieve the best results on complex mixed-type data, but it gives a point of reference and is simple to interpret (we can inspect reconstruction error, etc.). Also, many later GANs (like medBGAN, medWGAN) build on medGAN's concept, so it's instructive to include it.
2. **CTGAN (2019)** – *Conditional Tabular GAN* \cite{Xu2019CTGAN}: CTGAN is a state-of-the-art model for general tabular data. **Configuration:** We will use an open-source implementation (e.g., from the SDV library) of CTGAN. It employs a transformer to encode each column distribution and uses conditional vector sampling to ensure that each iteration focuses on a specific column's mode, thereby handling imbalanced categorical levels. The generator and discriminator are deep MLPs, and we will likely use default hyperparameters suggested by Xu et al. (like training epochs ~300, batch size ~500, generator lr ~2e-4, etc.). CTGAN can naturally handle mixed data by transforming continuous columns via a variational Gaussian mixture and one-hot encoding categoricals internally. **Rationale:** CTGAN is a strong baseline for any tabular data (health or otherwise). It often outperforms earlier models (like TableGAN or vanilla GANs) on benchmark datasets. We expect CTGAN to be a top-performing GAN in utility, so it's a fair yardstick to compare LLMs against. Additionally, CTGAN's design is fairly robust to a variety of tables, which is useful since we have multiple datasets.
3. **ADS-GAN (2020)** – *Anonymization through Data Synthesis GAN* \cite{ADS-GAN2020}: **Configuration:** We will implement ADS-GAN's approach by modifying a GAN (likely using CTGAN or WGAN-GP as a backbone) to include the identifiability loss. The identifiability loss requires computing distances between real and synthetic samples – to make this feasible, in each training batch, we'll generate a batch of synthetic samples and pair each with the closest real sample in that batch (or use all vs all). We'll then calculate a penalty term $L_{id} = \max(0, m - d(x_{\text{real}}, x_{\text{syn}}))$ for each pair, where d is a distance metric (like Euclidean) and m is a minimum margin we want to enforce. Summing this penalty over pairs gives the identifiability loss, which is added to the generator loss with some weight λ . Based on Yoon et al., we'll start with m as a quantile of real-real distances (ensuring typical real pairs are beyond m) and treat λ as a hyperparameter. The rest of ADS-GAN uses WGAN-GP which we will incorporate (gradient penalty on discriminator). **Rationale:** ADS-GAN is specifically chosen to test privacy-preserving GAN performance. It's one of the few models aimed at both privacy and utility. Including ADS-GAN allows us to see how a GAN with a privacy guard compares to an LLM with DP. It's configured to produce safe synthetic data by design, so if any model can match an LLM in privacy, this is a good candidate.
4. **DP-GAN (2018)** – (If time permits) a GAN trained with Differential Privacy \cite{Xie2018DPGAN}: **Configuration:** We would train, say, a CTGAN or WGAN model but apply DP-SGD to the discriminator. That means for each batch, we clip each per-sample gradient to a norm bound and add Gaussian

noise. We would decide an ϵ for training (like $\epsilon=1$ or 5) and tune the noise accordingly (with a given δ , maybe 10^{-5}). We might use the moments accountant to track privacy loss over epochs. The generator's updates don't access data directly, so they don't need noise. This yields a differentially private generator model at final. **Rationale:** This is a more formal way to ensure privacy than ADS-GAN's heuristic. It would be interesting to see how the utility compares: DP training can severely hurt the GAN's ability to converge if noise is high. If results are poor, that itself is a finding (perhaps explaining why not many works succeeded with DP-GANs). Due to complexity, this is an optional model; our main focus is ADS-GAN, but we include DP-GAN if resources allow.

LLM-Based Models:

1. **Pythia 160M / 1B (2023)** – *Pythia Large Language Models* \cite{Biderman2023}: We will use at least two model sizes from the Pythia suite, for example, a 160 million parameter model (which is reasonably small and trainable on a single GPU) and a 1 billion parameter model (which may require multi-GPU or gradient accumulation due to memory). **Configuration:** These models are GPT-style transformers (decoder-only architectures). We will fine-tune them on our datasets formatted as text. For instance, for MIMIC-III tabular data, each patient record might be converted to: "`<BOS> Age: 67; Gender: M; Diagnoses: Diabetes|Hypertension|...; Outcome: 1 <EOS>`" and tokenized. We will adapt the tokenizer to include needed tokens (like medical terms if they appear frequently). Fine-tuning will likely be done with a learning rate $\sim 1e-5$ or $1e-4$ (since starting from a pre-trained model, we don't want to distort it too heavily). We will run for a certain number of epochs or until convergence on validation perplexity. We will use teacher forcing (standard LM training). At generation time, we will prompt the model with a start token and have it generate until EOS token is produced. We might also try in-context learning: giving the model a few examples as prompt and then asking for a new sample (especially for smaller models if fine-tuning is not as effective). The 1B model will test the effect of a larger model capacity capturing more nuance in the data. **Rationale:** Pythia models were explicitly designed for research with transparency and come in multiple sizes, which is perfect for analyzing scaling effects. Also, Miletic & Sariyar used Pythia, so by doing the same, we can compare notes with their benchmark. The 160M vs 1B will allow us to see if larger truly is better in our setting (likely yes for utility, but possibly more risk for privacy if not DP-protected).
2. **GPT-2 Medium (2019)** – *OpenAI GPT-2* (around 345M parameters) \cite{Radford2019}: As an alternative LLM, or if Pythia infrastructure poses any issue, we can use GPT-2. **Configuration:** GPT-2 has pre-trained weights up to 1.5B, but the medium model (345M) is a good middle ground. Fine-tuning GPT-2 on our text data would be similar to Pythia. We might consider GPT-2 because it's a well-understood baseline and many libraries support it out of the box. **Rationale:** GPT-2 has been used in other synthetic data contexts (especially text generation). Having it gives us another point in the LLM landscape. We don't anticipate huge differences between GPT-2 and Pythia of similar size, but if there are, that could be insightful (perhaps differences in pre-training data might affect how well they model structured info). This is a secondary choice; if Pythia models suffice, we may not use GPT-2 to avoid redundancy.
3. **Differentially Private LLM (2024)** – *SafeSynthDP method applied to Pythia/GPT-2* \cite{Nahid2024}: **Configuration:** We will train one of the above LLMs with DP-SGD. Using Opacus or an equivalent library, we will set a clipping norm (e.g., 1.0) and noise multiplier to achieve desired ϵ after a given number of steps. Because language models have many parameters, DP training is

computationally heavy; we might focus on the smaller model (e.g., Pythia 160M or GPT-2 medium) for DP training. We'll experiment with a moderate privacy budget first (say $\epsilon=5$) to see if the model can still learn, then try stricter ($\epsilon=1$) if possible. We will monitor that the model isn't just outputting gibberish due to the noise. We might also implement data augmentation as suggested by some DP studies to reduce the effective sensitivity (like partitioning sequences). Another trick is to use **Noise at Generation**: after training a non-DP model, at generation time we sample from a slightly flattened probability distribution (adding entropy) which can serve as a form of privacy (though not formal DP). We might do that if DP-SGD proves too destructive. **Rationale:** This model is crucial for addressing RQ2 – we want to show how an LLM can be made privacy-preserving and measure its trade-offs. It will be directly compared to ADS-GAN or DP-GAN. We expect DP-LLM to have lower utility but significantly reduced memorization. If successful, it would underscore that LLMs can achieve strong privacy akin to DP GANs.

4. **Prompt-based LLM (GPT-4 or similar, 2023)** – *Large model via API* (optional exploratory): If resources and rules allow, we could use an API-accessible model like GPT-4 to generate some synthetic data by prompt engineering (e.g., prompt: "Generate 100 fictitious patient records with these fields..."). This wouldn't be a core part of our experiments since it's not easily controllable or reproducible without the API, and privacy is unclear (OpenAI's models aren't open). But it could be an interesting sanity check of what a very large model generates (sometimes GPT-3/4 can produce plausible data given a few examples). We mention it for completeness, but likely we will **not** rely on proprietary models for the main results.

Configuration Summary: We will ensure each model's configuration is optimized for our problem: - All models will output in a common format that we can parse into structured data for evaluation (for GANs it's already structured as vectors; for LLMs, we will parse the text). - For fairness, if doing conditional generation (e.g., class labels), we might give both GAN and LLM the same advantage (for example, telling them the class to condition on). This might not always be used, but if one approach benefits from conditioning (like CTGAN can condition on a class label), we could also prompt the LLM with that class info to generate accordingly. - Random seeds and multiple runs will be done to account for variability, especially for GANs.

Rationale Recap: By choosing medGAN, CTGAN, and ADS-GAN, we cover the GAN evolution from early to state-of-art with privacy; by choosing Pythia/GPT-2 with and without DP, we cover LLM evolution and privacy. This ensemble of models will allow us to answer which class of models is better and under what circumstances. If one specific model shines (say ADS-GAN or DP-Pythia), we can drill down into why.

We also note that implementing and running all these is a lot of work, so we will prioritize as needed: CTGAN vs Pythia (non-DP) is highest priority for utility comparison; ADS-GAN vs DP-Pythia is highest for privacy comparison. Others (medGAN, DP-GAN, etc.) add context and completeness but are secondary.

Evaluation Metrics

We will employ a comprehensive set of **evaluation metrics** to rigorously assess the synthetic data generated by each model, focusing on both privacy preservation and data utility. These metrics will allow quantitative comparisons and trade-off analysis.

Privacy Metrics:

- **Membership Inference Attack (MIA) Success Rate / AUC:** This metric evaluates how well an adversary can guess if a particular data record was in the training set of the generative model. We will measure the success of membership inference in two ways: (1) as an accuracy (or true positive rate at a given false positive rate), and (2) as the area under the ROC curve (AUC) for the adversary's classifier. For example, we might use the distance-to-nearest-synthetic neighbor as a test statistic for each real record, and choose a threshold: records with very close synthetic neighbors are classified as "in training set". We can compute the true positive rate (TPR) and false positive rate (FPR) of this classification by testing on known members vs non-members. By varying the threshold, we get an ROC curve and compute AUC. If AUC = 0.5, the synthetic data provides no advantage to the adversary over random guessing (good for privacy); if AUC is significantly >0.5 , there's information leakage. We will report this for each model. We expect that models without privacy guard might have AUC somewhat above 0.5, while models with strong privacy (DP) should be ~ 0.5 . This directly answers how vulnerable each model is to membership inference \cite{Shokri2017}.
- **Attribute Disclosure Risk / Precision:** This metric assesses whether an adversary can correctly infer hidden attributes of individuals using synthetic data. We will simulate an adversary who knows some attributes (say a subset of features) of a person and tries to predict an unknown sensitive feature by looking at synthetic data. We quantify risk as the probability that the adversary's guess is correct, compared to a baseline probability (like prevalence of that attribute). One way to measure this: for each real individual (in a hold-out set), hide one attribute's value. Then, using synthetic data, have the adversary find records that match the known attributes and use, for instance, a majority vote or probability estimate from those synthetic records to predict the missing attribute. Compute the **precision/recall** of these guesses over many trials. We will specifically focus on sensitive attributes (e.g., presence of a particular diagnosis). A high precision (significantly above baseline) indicates a privacy issue. We can condense this into a single number like "precision at one guess" or average information gain. Another approach is *Mutual Information*: measure if synthetic data increases mutual information between known and sensitive attributes from the adversary's perspective. But to keep it simple, we'll likely report something like: *adversary success rate in guessing attribute = X% vs baseline Y%*. This metric complements membership inference by focusing on partial information leakage.
- **Identifiability Score / Nearest Neighbor Distance:** We will calculate, for each synthetic record, the distance to its nearest real record (in the training set). Then we look at the minimum such distance (worst-case scenario) and the distribution (perhaps 5th percentile). If any synthetic record is extremely close (below a certain distance threshold) to a real one, that synthetic record might be essentially revealing that real record. We can define an **identifiability score** as the fraction of synthetic records that are within δ distance of some real record (for some small δ we define). Alternatively, report the min distance found. A higher min distance is better (means the model didn't just reproduce a real point). We might use Hamming distance for binary vectors (medGAN case) or Euclidean for continuous. For mixed data, we may use a normalized distance metric (e.g., treat each feature distance normalized by range, then L2 norm). Yoon et al.'s identifiability metric essentially used this concept \cite{ADS-GAN2020}. We will present either a plot or summary of these distances. This is an intuitive privacy metric: for example, "In the vanilla GAN synthetic data, 2% of synthetic records had a near-identical match in the real data (distance < 0.1), whereas in ADS-GAN or DP-LFM, 0% did."

- **Differential Privacy Budget (ϵ) and Theoretical Guarantees:** For models where we apply formal differential privacy, we will report the (ϵ, δ) values achieved. For instance, "This LLM was trained with $(\epsilon=2, \delta=10^{-5})$ -DP". While ϵ itself is not an empirical metric, it's a parameter that gives an upper bound on privacy loss. We will discuss it to contextualize the empirical metrics. For example, if a model has $\epsilon=1$, we expect membership inference to be nearly impossible; if a model has no formal DP but still yields good empirical privacy, that's noteworthy too. So ϵ will appear in our analysis for the DP-trained models as a metric of privacy guarantee strength.

Utility Metrics:

- **Classification/Prediction Accuracy (or AUROC/F1):** The main utility metric will be the performance of ML models trained on synthetic data on a real-data test. Depending on the task (binary classification, multi-class, regression), we will use appropriate metrics: accuracy or AUROC for classification, F1-score for imbalanced binary outcomes, or mean squared error for regression tasks (like maybe predicting length of stay as a continuous outcome). Specifically, for each generative model's synthetic dataset, we will train a classifier (we might use a standard algorithm like Random Forest or a simple neural network) and evaluate on the real test set labels. We will likely present this as a percentage of the real-data-trained model's performance. For example: *Model trained on real data achieves 0.85 AUROC; model trained on GAN synthetic achieves 0.80 AUROC (94% of real performance); model trained on LLM synthetic achieves 0.83 (98% of real)*. Such comparisons succinctly measure utility \cite{Choi2017, ADS-GAN2020, Miletic2024}. We might also do the reverse: train on real, test on synthetic as a measure of distribution overlap (ideally, a classifier should perform similarly in both directions if distributions match). We will report results for key tasks like mortality prediction or disease classification, whichever tasks we set up for each dataset.
- **Statistical Similarity (Distributional Metrics):** We will evaluate how closely the synthetic data distribution matches the real data distribution. This includes:
 - *Marginal Distributions:* For each feature, we can compute statistical distances. For continuous features, we could use **Wasserstein Distance (Earth Mover's)** or **Kolmogorov-Smirnov (KS) statistic** comparing real vs synthetic. For categorical features, we can use **Jensen-Shannon divergence** between the real and synthetic frequency distributions. We will likely present an average divergence across features, or highlight specific important features. If the dataset is large, even a small JS divergence can be significant, so we might run hypothesis tests like KS test and report whether the null (synthetic same as real) is rejected or not for each feature. Ideally, a good generative model will have most features with no significant difference in distribution.
 - *Pairwise Correlations:* We will compute the correlation matrix for real data and for synthetic data (for numeric features, use Pearson or Spearman; for categorical combinations, use Cramer's V or similar). Then we can compute an error metric like the Frobenius norm of the difference between correlation matrices, or simply report a few key correlation comparisons. For example, if in real data age and blood pressure have correlation 0.3, but in synthetic it's 0.1, that indicates a utility loss in capturing that relationship. A more advanced metric some use is the **Maximum Mean Discrepancy (MMD)** over the joint distribution, but that might be overkill for our needs. Instead, summarizing one and two-dimensional distributions should suffice.

- **Multivariate Overlap:** As mentioned, we may train a logistic regression to discriminate real vs synthetic (assign label 0 to real samples, 1 to synthetic, and see how well it can classify). If it has an AUC of 0.5 (random guessing), the distributions are very similar. If it's high, there's a detectable difference. This has been used as a metric in some works (a low detection rate is good as a "Giant Turing test"). We can report the accuracy of this *real-vs-fake classifier* as a summary measure of distribution overlap. A value near 50% is ideal for utility (though one must be cautious because an extremely noisy synthetic data might also appear "different" and thus be classified easily as fake, so it's not purely monotonic with quality).
- **Downstream Task-Specific Metrics:** If we consider any task beyond classification, e.g., for text data might be BLEU score or ROUGE if comparing synthetic text to real text patterns, or for sequential data maybe some clustering measure. For the structured data, classification is main; if we had a regression (like predicting a lab value), we could use R^2 or MSE. We will list such metrics if relevant to a dataset. For example, in the heart disease dataset (which is classification), accuracy and ROC is fine. In a survival analysis context, maybe C-index could be used if we tried that. We likely will stick to classification metrics.
- **Quality of Fit for Models Retrained on Synthetic:** Another viewpoint: take a model trained on real data (like a simple linear regression on real), evaluate it on synthetic data (for consistency) and vice versa. This checks if patterns learned on one carry over. If a model's coefficients are similar when trained on real vs synthetic, that's a utility sign too. We might not have time to go deep into this, but it's another analysis angle.

Trade-off Evaluation: We will explicitly combine the above metrics to illustrate trade-offs: - For example, we may create a plot with x-axis as a privacy metric (like membership inference AUC or epsilon value) and y-axis as a utility metric (like classification accuracy). Each model or each setting (like different DP noise levels) will be a point. This will show the privacy-utility frontier. We will highlight which model yields the best trade-off (Pareto optimal front). If, say, ADS-GAN and DP-LLM points are above others, those are preferred methods.

We will also evaluate metrics in intermediate regimes (e.g., ADS-GAN with different identifiability loss weights, or LLM with various ϵ) to see the curve. Possibly something like: a table of (Utility, Privacy) for each method:

Model	Accuracy (% of real)	MI Attack AUC	Attr Inf precision
Identifiability minDist			
CTGAN	95%	0.65	0.50 (baseline 0.50)
0.02			
ADS-GAN	93%	0.55	0.51
			0.10
Pythia LLM	97%	0.60	0.52
			0.03
DP-Pythia LLM	90%	0.50	0.50
			0.08
...			

This hypothetical table would be insightful.

Beyond numbers, we will interpret them: e.g., if an MI attack AUC is 0.6, that means the adversary has a 60% chance to distinguish members vs non-members. We'd call that moderate leakage. If identifiability

minDist is 0 (some synthetic = real exactly), that's a severe breach. These interpretations will be in the discussion.

Finally, we plan to engage in some *statistical testing* on these metrics. For instance, to claim one model's synthetic data leads to significantly better accuracy than another's, we might train and evaluate multiple times and do a t-test on the results. Similarly for privacy metrics: we can get many samples of an attack success (by randomly picking target records) and test differences. If differences are large, formal tests may not be needed; but we will mention if differences are statistically significant or just observed.

In conclusion, these metrics collectively ensure that we rigorously quantify what we mean by "utility" and "privacy". They provide different perspectives: predictive performance, distributional fidelity, membership leakage, attribute leakage, and so on. By examining all, we will have a well-rounded assessment of each approach. Our thesis will present these results in a clear manner (graphs, tables) and use them as evidence to answer the core questions. For example, if LLM has higher accuracy but also higher MI risk than a certain GAN, we can explicitly point that out, and the decision of "which is better" can be nuanced depending on whether utility or privacy is paramount in a given application.

Timeline

The project is planned to span **6 months (approximately 24 weeks)**. Below is a month-by-month breakdown of the major phases and milestones:

- 1. Month 1 – Literature Review and Proposal Refinement:** Conduct an in-depth review of related work in synthetic data generation, privacy-preserving machine learning, and healthcare data sharing. Read and summarize key papers (GAN variants, DP techniques, LLM synthetic data papers) beyond those already gathered. Simultaneously, refine the thesis proposal and research plan (this document) based on feedback and new insights. By the end of Month 1, finalize the specific research questions, the experimental plan, and ensure access to all needed resources (e.g., obtain MIMIC-III access approval, set up computing environment for training large models).
- 2. Month 2 – Dataset Preparation and Baseline Experiments:** Acquire and preprocess the chosen datasets (MIMIC-III subsets, Heart Disease, etc.). This includes cleaning data, selecting relevant features, splitting into train/validation/test sets, and formatting data for model training (e.g., text serialization for LLMs, normalization for GANs). Begin with baseline experiments: train a simple model on real data to record baseline predictive performance for each task (this helps later to gauge synthetic utility). Also, if possible, run a quick test of a basic GAN (maybe CTGAN) and a basic LLM (small GPT-2) on one dataset with a few epochs to identify any immediate issues (such as format problems or model instabilities). These dry-run experiments will ensure the pipeline is functioning. By end of Month 2, we should have all data ready and have initial results from baseline models on real data.
- 3. Month 3 – GAN Model Implementation and Training:** Implement or set up the GAN-based models (medGAN, CTGAN, ADS-GAN). This includes coding custom loss functions (identifiability for ADS-GAN, etc.) and verifying that the implementations work on a toy example. Train each GAN model on each dataset's training portion. This will likely involve multiple runs to tune hyperparameters. Use the validation set to tune things like the identifiability loss weight or DP noise level (if using DP-GAN). Monitor training curves, adjust as needed for stability. By mid-Month 3, have at least one well-

trained model for each GAN variant on the primary dataset (MIMIC-III, for instance). By end of Month 3, generate synthetic datasets using these GANs and perform preliminary evaluation (distribution stats, maybe a quick utility check) to ensure the outputs make sense (e.g., no obvious mode collapse or all-same outputs). If any GAN model fails (e.g., mode collapse that can't be fixed), decide whether to drop it or replace it with another approach.

4. **Month 4 – LLM Model Implementation and Training:** Set up the LLM fine-tuning environment. This may involve using GPU clusters due to model sizes. Fine-tune the chosen LLMs (Pythia models of two sizes, GPT-2 etc.) on the datasets. This will be computationally heavy; plan training runs carefully (maybe start with the smaller model and one dataset to estimate time, then scale up). Also implement the DP-SGD training for LLM for at least one configuration. Month 4 might be iterative: train non-DP LLMs first (to get utility results), then train DP versions (which might require more epochs or different hyperparams). Keep an eye on validation loss to avoid overfitting. By mid-Month 4, have the non-DP LLM synthetic data generated. By end of Month 4, ideally have the DP-LFM model trained or at least in progress (DP might require more epochs so could spill into Month 5). During this month, also start rigorous evaluation of the GAN results from Month 3: run the full evaluation metrics on those and store the results.
5. **Month 5 – Comprehensive Evaluation and Analysis:** This month focuses on evaluating all models' outputs and analyzing the results. Generate synthetic datasets for all trained models (GANs and LLMs, DP and non-DP). Perform the utility evaluations: train classifiers on synthetic data, evaluate on real test sets, record accuracies/AUCs for each combination. Perform the privacy evaluations: run membership inference experiments, attribute inference simulations, compute identifiability metrics, etc., as described in the methodology. This will involve writing some attack code and running it multiple times for statistical significance. Organize the results into tables/plots. By mid-Month 5, have a clear comparative picture of at least the main scenarios (e.g., CTGAN vs LLM on MIMIC, AD-SGAN vs DP-LLM on MIMIC, and similar for other datasets). In the latter half of Month 5, interpret the results: identify patterns (e.g., "LLM outperforms on utility but leaks more on X metric", or "with $\epsilon=1$, both methods suffer utility drop of Y% but privacy is nearly perfect"). Possibly perform additional experiments if needed to clarify questions that arise (for instance, if results are surprising, run a confirmatory experiment or adjust parameters to see sensitivity). Also, this month, if any dataset or model was left (like text data generation if planned), attempt it and do a qualitative evaluation.
6. **Month 6 – Writing and Finalization:** Begin writing the thesis document (chapters on methodology, experiments, results, discussion) in an Overleaf project. This includes describing the methods (much of which is in this proposal but with more implementation details and perhaps equations), documenting results, and creating figures and tables from Month 5's analysis. Simultaneously, reflect on the findings and formulate the discussion: what do the results mean for researchers or practitioners? What are the limitations of this study (e.g., maybe results might differ on other data not tested, or the computational cost factor)? Also plan the conclusion and possibly future work suggestions. Around mid-Month 6, have a full draft of the thesis ready for review by advisors. Incorporate feedback and revisions in the second half of the month. Prepare any additional materials required for defense (like slides, if a defense is scheduled soon after). By the end of Month 6, the thesis should be finalized, well-formatted (with all citations, figures, etc.), and ready for submission. The conclusion and abstract will be refined to accurately summarize the comparative insights gained.

Throughout the timeline, we will have regular meetings with the thesis supervisor to report progress and adjust plans as necessary. We also budget some buffer time within each month for unforeseen delays (e.g., a model taking longer to train, or needing to troubleshoot a bug in code, etc.). If everything goes smoothly, any spare time will be used to either explore an additional angle (like using the synthetic data to train a different kind of model, or trying a different privacy attack) or to polish the presentation of results.

By following this timeline, we ensure steady progress from groundwork (literature, data prep) to execution (model training) to synthesis (evaluation, writing). Each step builds towards the ultimate goal: a thorough and well-supported thesis answering how GANs vs LLMs compare for privacy-preserving synthetic data in healthcare.

Conclusion

In this thesis, we will deliver a detailed comparative study of **GAN-based vs. LLM-based synthetic data generation in healthcare**, with a focus on the crucial balance between data utility and privacy preservation. The healthcare domain presents unique challenges – sensitive personal information is entwined with complex, high-dimensional data needed for life-saving machine learning applications. By exploring two cutting-edge generative strategies side by side, our study will provide insights into which approach (or combination) is better suited for different scenarios under the stringent requirements of health data sharing.

We anticipate several key contributions and conclusions from this work: - We will quantify the extent to which *large language models* (like transformer-based generative models) can reproduce the utility of real clinical datasets, potentially outperforming traditional *generative adversarial networks* on certain metrics. If our results align with recent findings \cite{Miletic2024}, we might conclude that LLMs are not only viable but perhaps *preferable* for synthetic tabular data generation, given enough data and model capacity. - Conversely, we will assess whether GANs, which have a longer track record in this domain, hold advantages in terms of *privacy*. Through membership and attribute inference tests, we will determine if the adversarial training in GANs inherently offers some privacy (e.g., by not explicitly storing training samples), or if LLMs with their memorization tendency pose greater privacy risks unless mitigated by additional techniques. We expect to demonstrate that privacy-focused GAN variants like ADS-GAN and DP-enhanced generative models (GAN or LLM) can drastically reduce identifiable information in the synthetic data. - A major outcome will be the **characterization of the privacy-utility trade-off** for each approach. For instance, we might find that to achieve a given level of privacy (say no higher than 0.55 AUC on membership inference), the LLM might incur a larger drop in utility than the GAN, or vice versa. These findings will be practically useful: a healthcare organization could decide which method to use based on what they prioritize. If our study finds that "Method A yields 90% of model accuracy with strong privacy, whereas Method B yields 95% accuracy but weaker privacy", this is exactly the nuanced information stakeholders need to make informed decisions. - Our literature review and experiments will also highlight **methodological considerations**: how to evaluate synthetic data quality comprehensively, what metrics are most indicative of success, and where current methods fall short. For example, we might conclude that while accuracy on downstream tasks is preserved, certain subtle statistical biases might creep in (perhaps one method fails to capture rare disease co-occurrences, which could be critical for specific research questions). Recognizing such limitations will suggest areas for future improvement (like incorporating domain knowledge or constraints into generation). - We will also discuss **computational and practical aspects**: LLMs require more computational resources – is the improved performance worth the cost in a typical research setting? GANs can be more finicky to train – does that limit their usability in practice? These real-world factors will temper our

conclusions with an understanding of feasibility. For instance, if we find an LLM approach needs a billion-parameter model to beat a GAN, a hospital with limited ML infrastructure might lean towards the simpler GAN solution for now. - Importantly, our conclusion will emphasize that the goal is not to declare an absolute "winner" but to provide guidance. It may turn out that **both approaches have merit**: GANs might be well-suited for certain types of data (or where explicit privacy guarantees are needed via DP), whereas LLMs might excel when data is plentiful and one needs very high fidelity. We will articulate these contexts. For example, perhaps for generating synthetic free-text clinical notes, LLMs are the obvious choice (since GANs can't do that effectively), whereas for small structured datasets, a carefully regularized GAN might outperform an LLM that would otherwise overfit.

In conclusion, this study will advance the state of knowledge in synthetic data generation for healthcare by providing an **empirical, head-to-head comparison** of two prominent methodologies under a rigorous evaluation framework. Our findings will inform data scientists, healthcare IT professionals, and policymakers about the strengths and weaknesses of GANs vs. LLMs for creating shareable, privacy-preserving datasets. By doing so, we hope to contribute to enabling more widespread data sharing and collaboration in healthcare, bridging the gap between privacy requirements and the data needs of modern AI.

Ultimately, the thesis will not only answer the research questions posed but also serve as a reference for best practices in evaluating synthetic data: illustrating how multiple metrics can be used together and how different privacy-preserving techniques compare. The insights gained could spark further research – for instance, if LLMs are shown to be very effective, one might explore hybrid models (combining GAN discriminators with LLM generators) or new regularization techniques to harness the best of both worlds. If GANs are shown to still be competitive, that validates continuing research into GAN improvements in the tabular domain.

The hope is that by illuminating the trade-offs, we empower stakeholders to make evidence-based decisions on deploying synthetic data solutions, thereby accelerating healthcare AI development in a safe and compliant manner. The successful completion of this project will be an academically rigorous addition to the privacy-preserving data generation literature and a practical guide for the community interested in synthetic health data.
